DECK 8 OF 10 · FRC JAVA CRASH COURSE

# Understanding motors: read the curve

Why your robot speeds up, slows down, heats up and browns out

WPILib 2026 · Java · Command-based

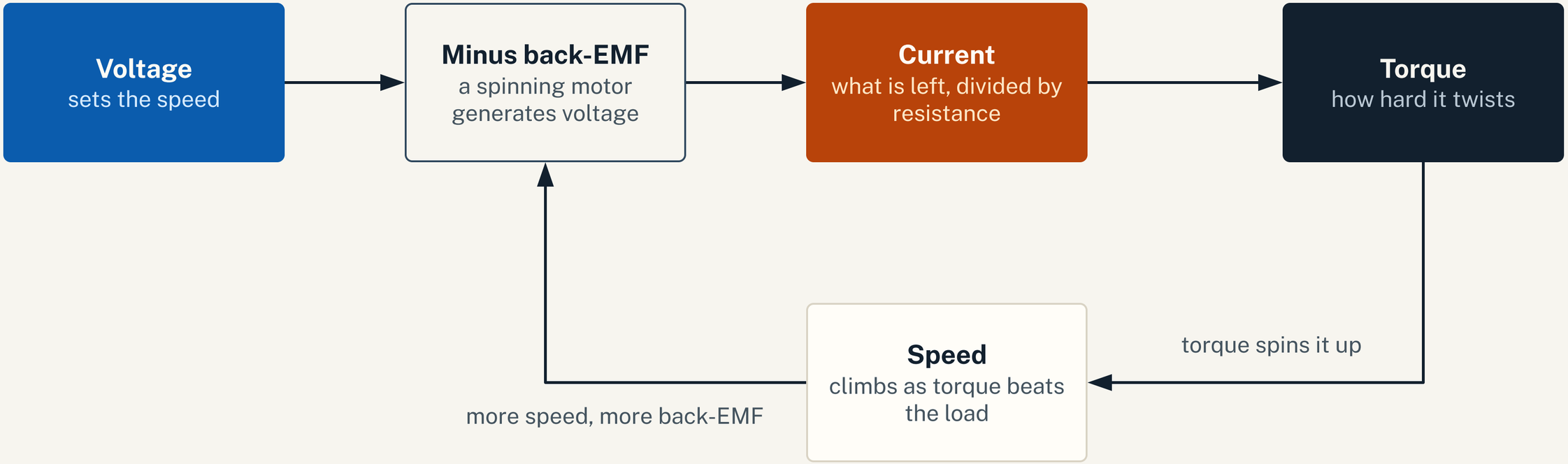
OVERVIEW

# What this deck covers

- What **voltage** does to a motor, and what **current** does.
- Free speed, free current, stall torque and stall current on a motor curve.
- Working out a robot’s top speed from its gearing, by hand.
- Why a stalled motor heats up, and why the battery sags.
- **Motor current** versus **battery current**, and choosing a current limit.

MENTAL MODEL

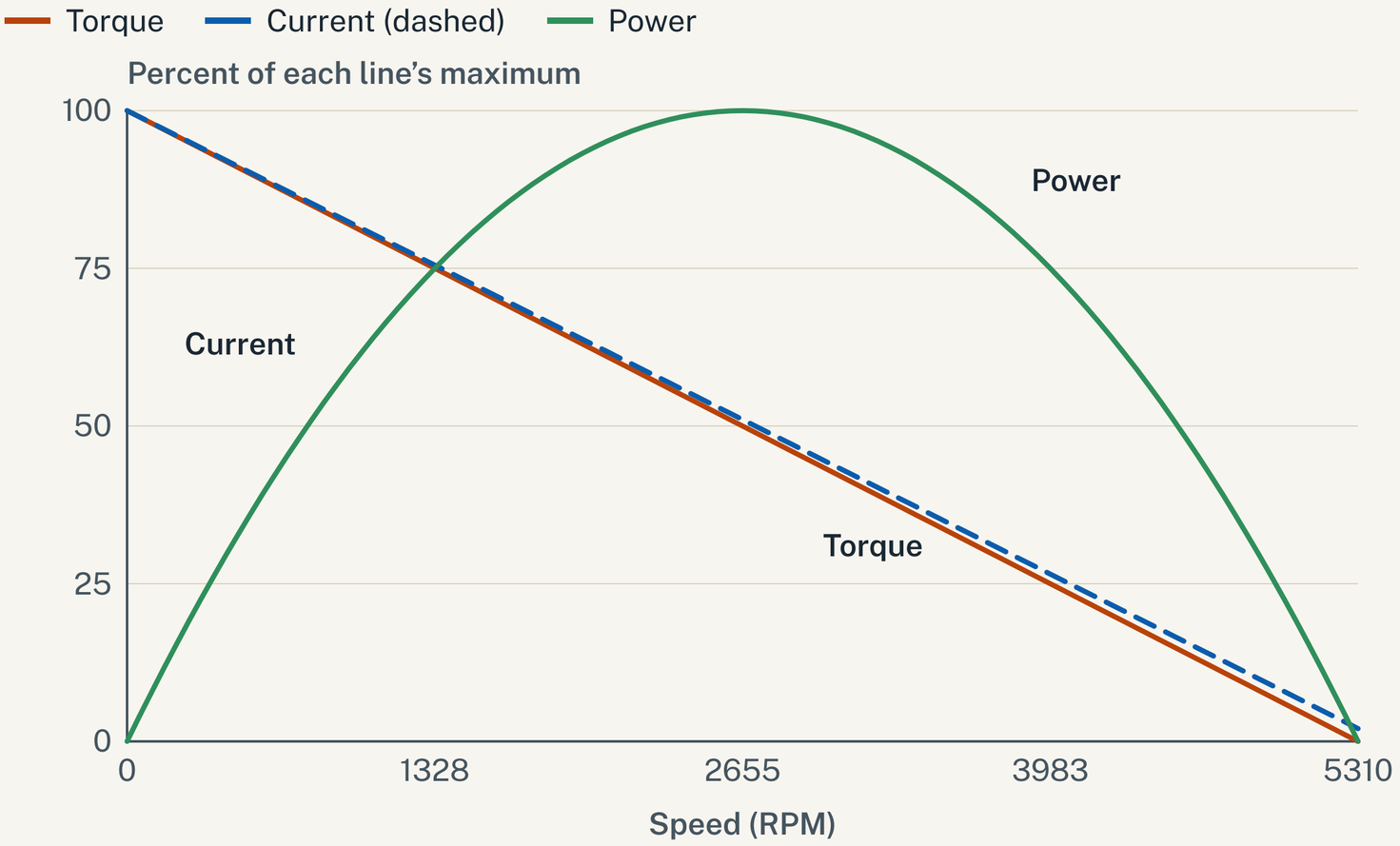
# Voltage sets speed, current sets torque




Schematic. The loop settles when back-EMF leaves just enough current to carry the load.

MOTOR CURVES

# Anatomy of a motor curve: a CIM at 12 V




1. **Stall: left edge**
   Stuck at 0 RPM: **2.42 Nm** of torque and **133 A**.

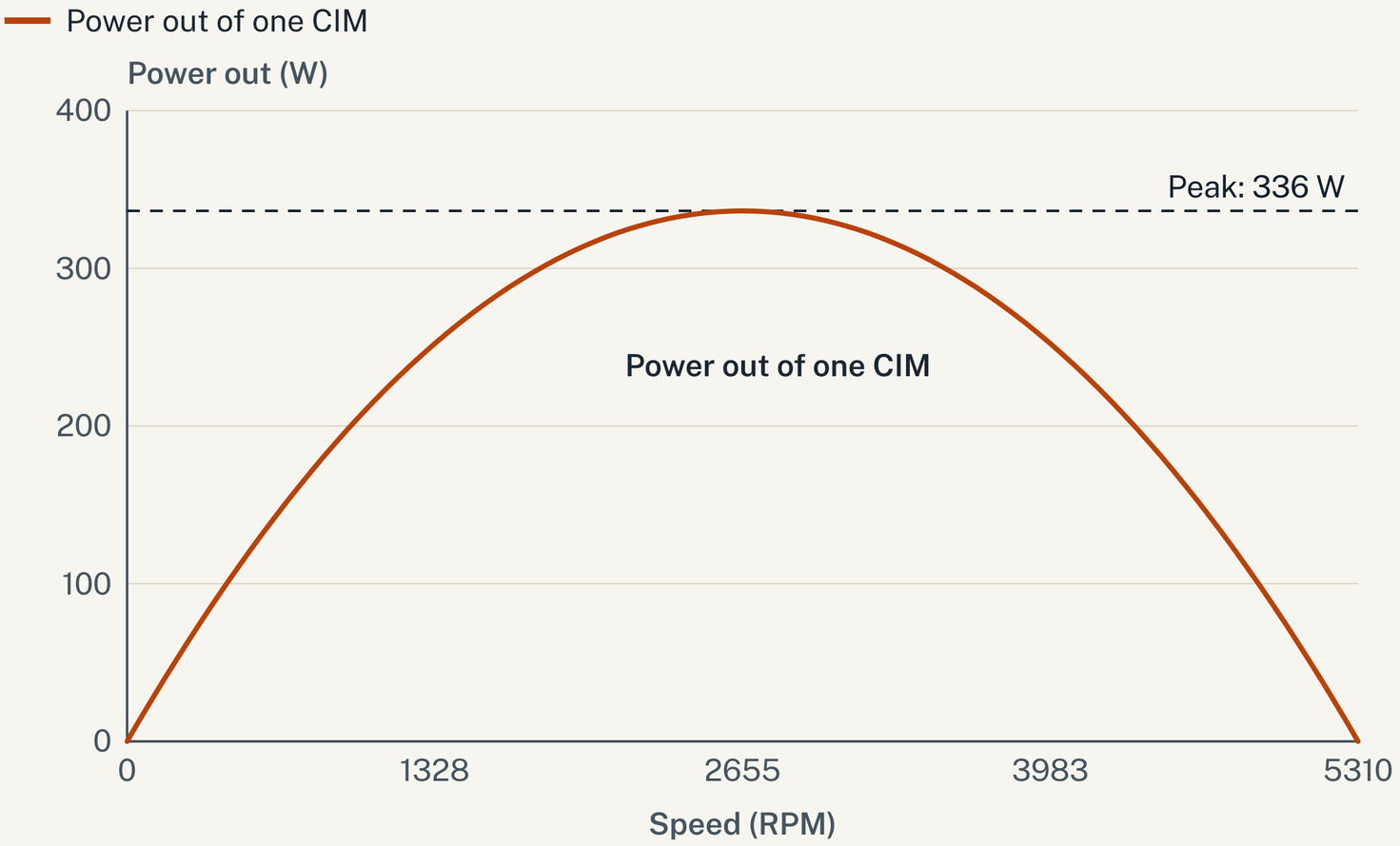
2. **Free speed: right edge**
   **5310 RPM** with nothing attached, and no torque left.

3. **Free current**
   **2.7 A** at free speed. That just beats friction.

4. **Two straight lines**
   Current makes torque, so the two fall together.

MOTOR CURVES

# Peak power sits at half speed




1. **The formula**
   Power = torque × speed, with speed in radians per second (RPM × 0.105).

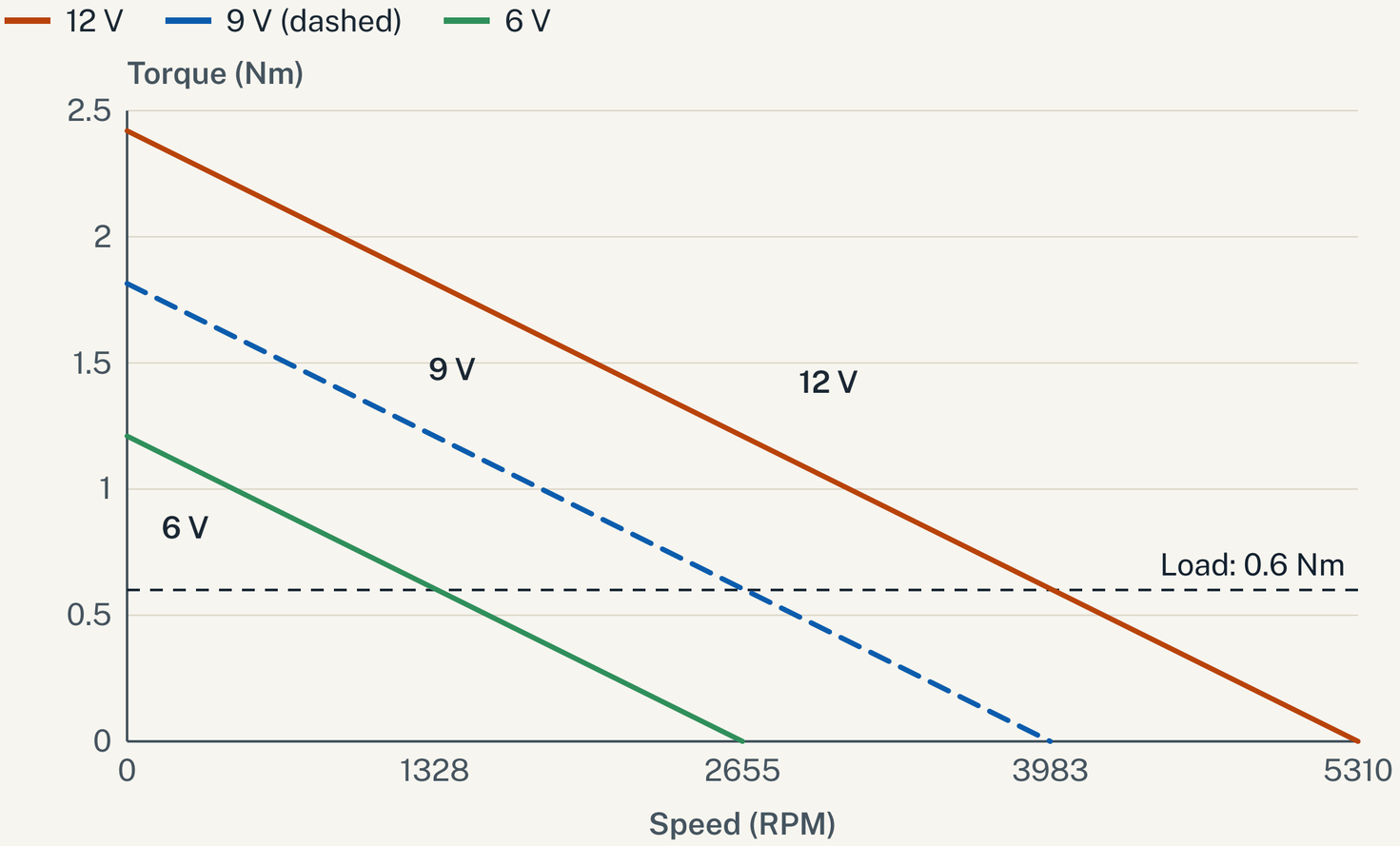
2. **Stall**
   All torque, no speed: 0 W.

3. **Free speed**
   All speed, no torque: 0 W.

4. **Half speed**
   Half of each: 1.21 Nm × 278 rad/s = **336 W**.

CONTROL THEORY

# Voltage slides the curve, load picks the point




1. **Your stick**
   set(0.5) on a 12 V battery gives the motor 6 V.

2. **Same slope**
   Less voltage moves the line down. Its shape stays.

3. **Load picks the point**
   The motor runs where the lines cross: 3993 RPM at 12 V, 1338 RPM at 6 V.

4. **Same torque, same amps**
   That 0.6 Nm takes 33 A on all three lines.

MOTOR CURVES

# Reading real numbers: four FRC motors

| Motor | Free speed (RPM) | Free current (A) | Stall torque (Nm) | Stall current (A) | Peak power (W) |
|---|---|---|---|---|---|
| CIM | 5310 | 2.7 | 2.42 | 133 | 336 |
| NEO | 5676 | 1.8 | 2.6 | 105 | 386 |
| NEO Vortex | 6784 | 3.6 | 3.6 | 211 | 639 |
| Kraken X60 | 6000 | 2 | 7.09 | 366 | 1114 |

From WPILib DCMotor at v2026.2.1. The Kraken row is the non-FOC entry. Peak power here is stall torque × free speed ÷ 4.

GEARING

# Gearing trades speed for torque: the KitBot

1. **Start at the motor**
   5310 RPM free speed and 2.42 Nm stall torque.

2. **Divide speed by the ratio**
   5310 ÷ 8.45 = **628 wheel RPM**.

3. **Multiply torque by the ratio**
   2.42 × 8.45 = **20.4 Nm** per CIM at the wheels.

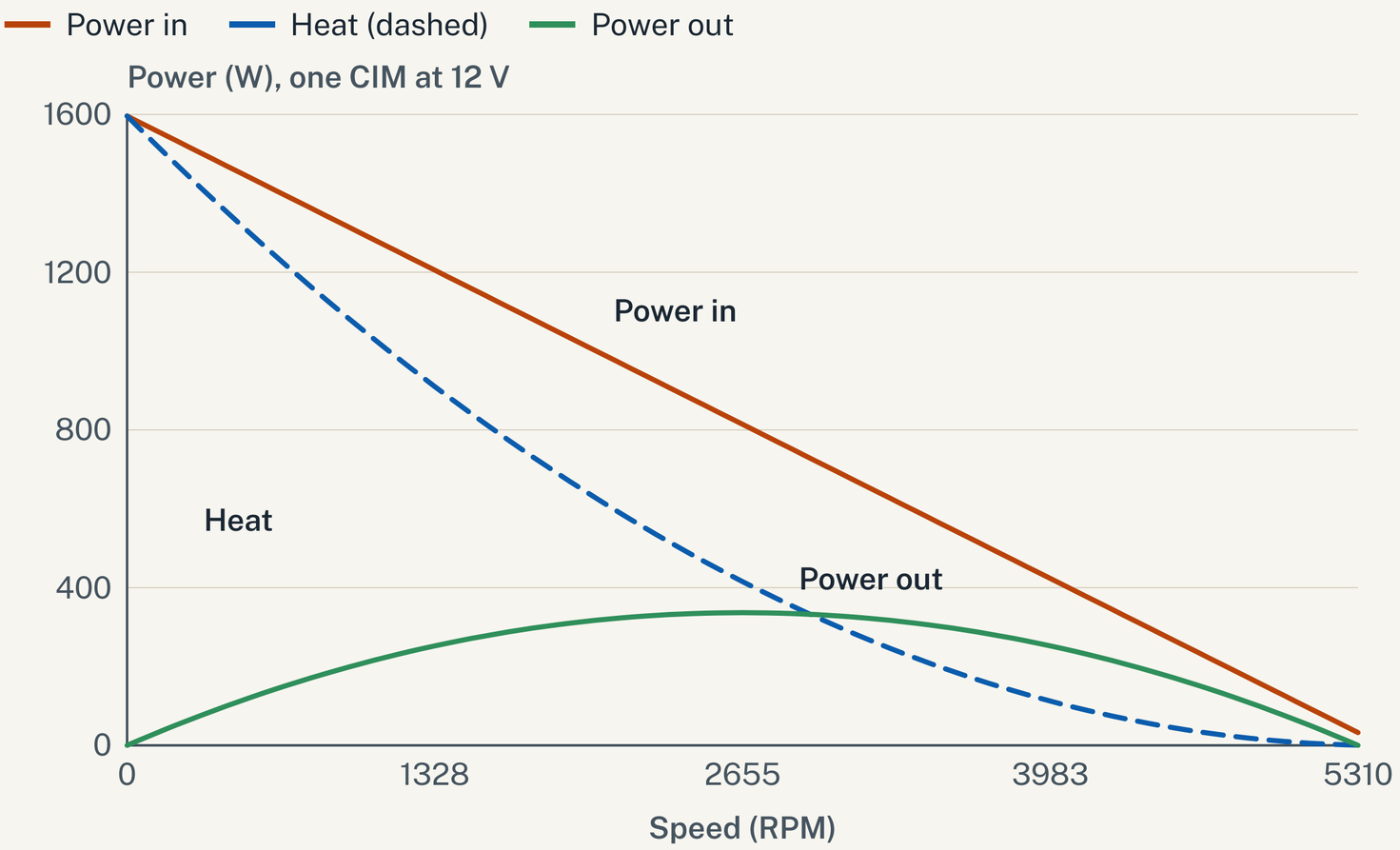
4. **One wheel turn rolls one circumference**
   6 in × π = 18.85 in, which is 0.479 m.

5. **Turns per second × distance per turn**
   628 ÷ 60 × 0.479 = **5.0 m/s** (16.5 ft/s).

6. **Compare with the real robot**
   FIRST quotes ~12.7 ft/s (3.87 m/s), 77% of this free-speed math.

HEAT

# A stalled motor turns all its power into heat



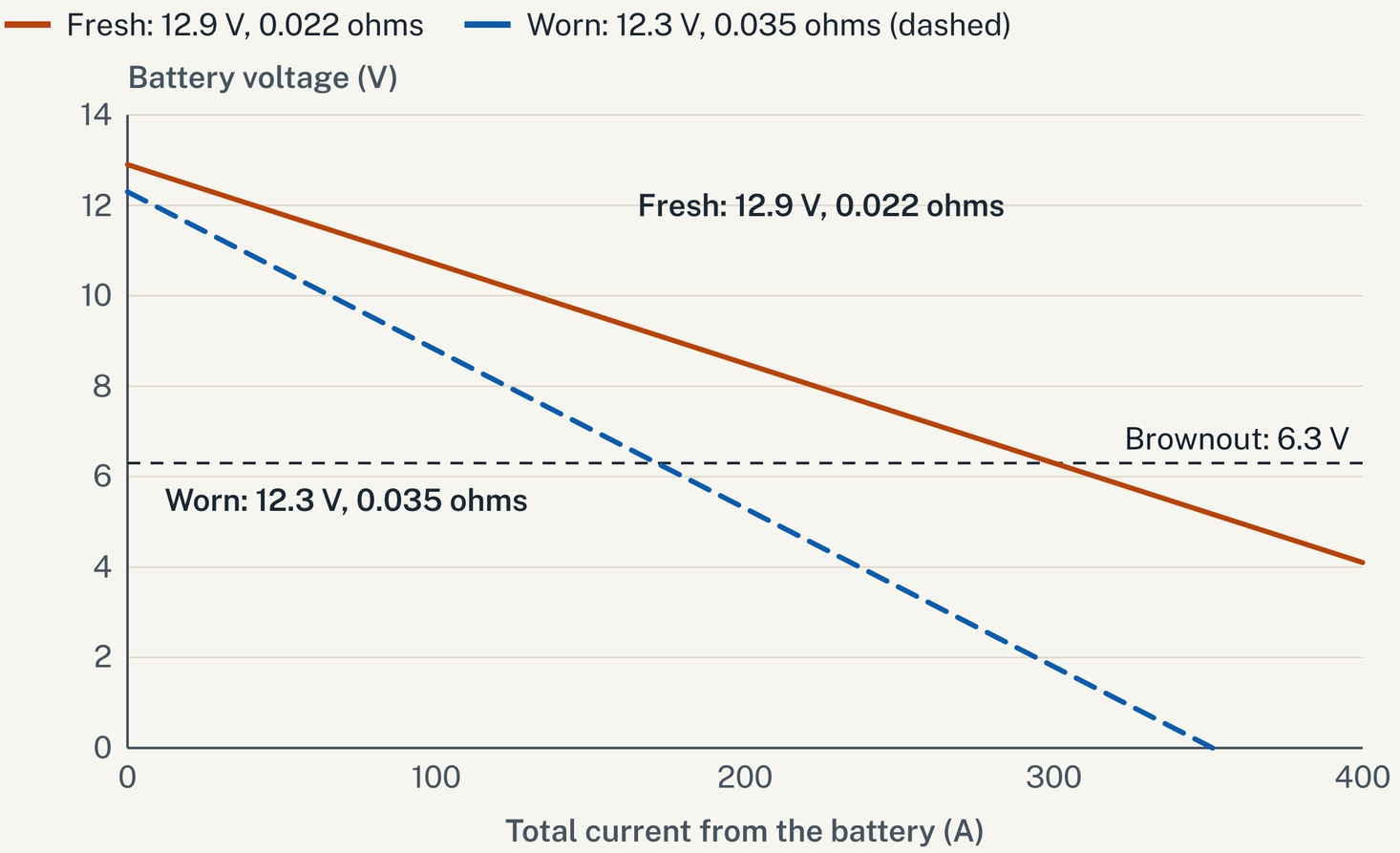

1. **At stall**
   12 V × 133 A = **1596 W** in, and nothing out.

2. **Where it goes**
   No back-EMF, so only the resistance of the windings limits the current. They heat up.

3. **At half speed**
   814 W in, 336 W out: 41% useful.

4. **Current squared**
   Half the current means a quarter of the heat.

# Battery sag and brownouts




1. **Sag**
   Every amp drops voltage inside the battery.

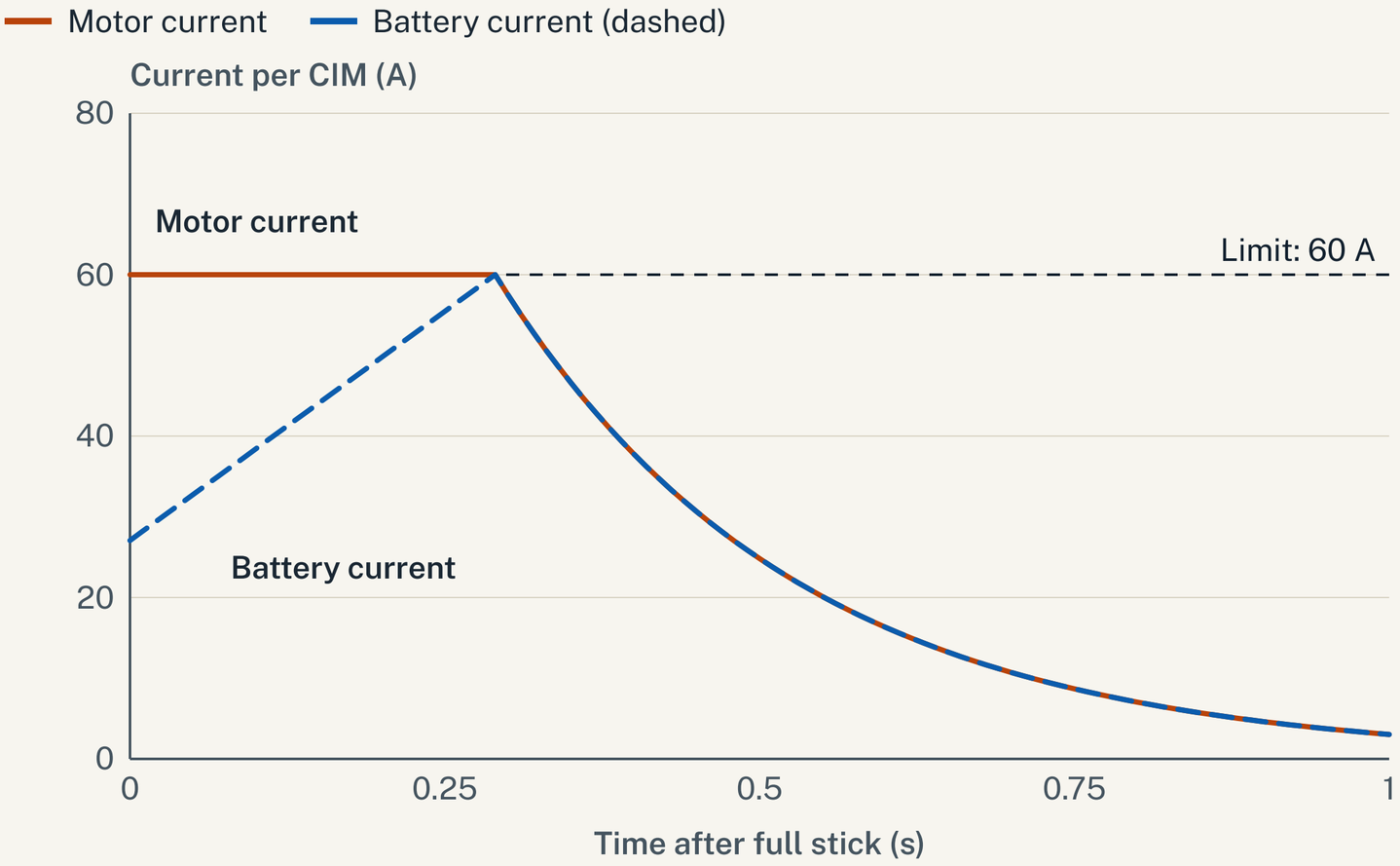
2. **Brownout**
   Below 6.3 V a roboRIO 1 cuts the motors. A roboRIO 2 defaults to 6.75 V.

3. **Full stick from a stop**
   No limit, fresh battery: 290 A, down to 6.5 V.

4. **Breakers**
   120 A main, 40 A per motor. They trip on heat after a delay, so spikes get through.

CURRENT

# Motor current and battery current are different




1. **Motor current**
   In the windings. It makes torque and heat.

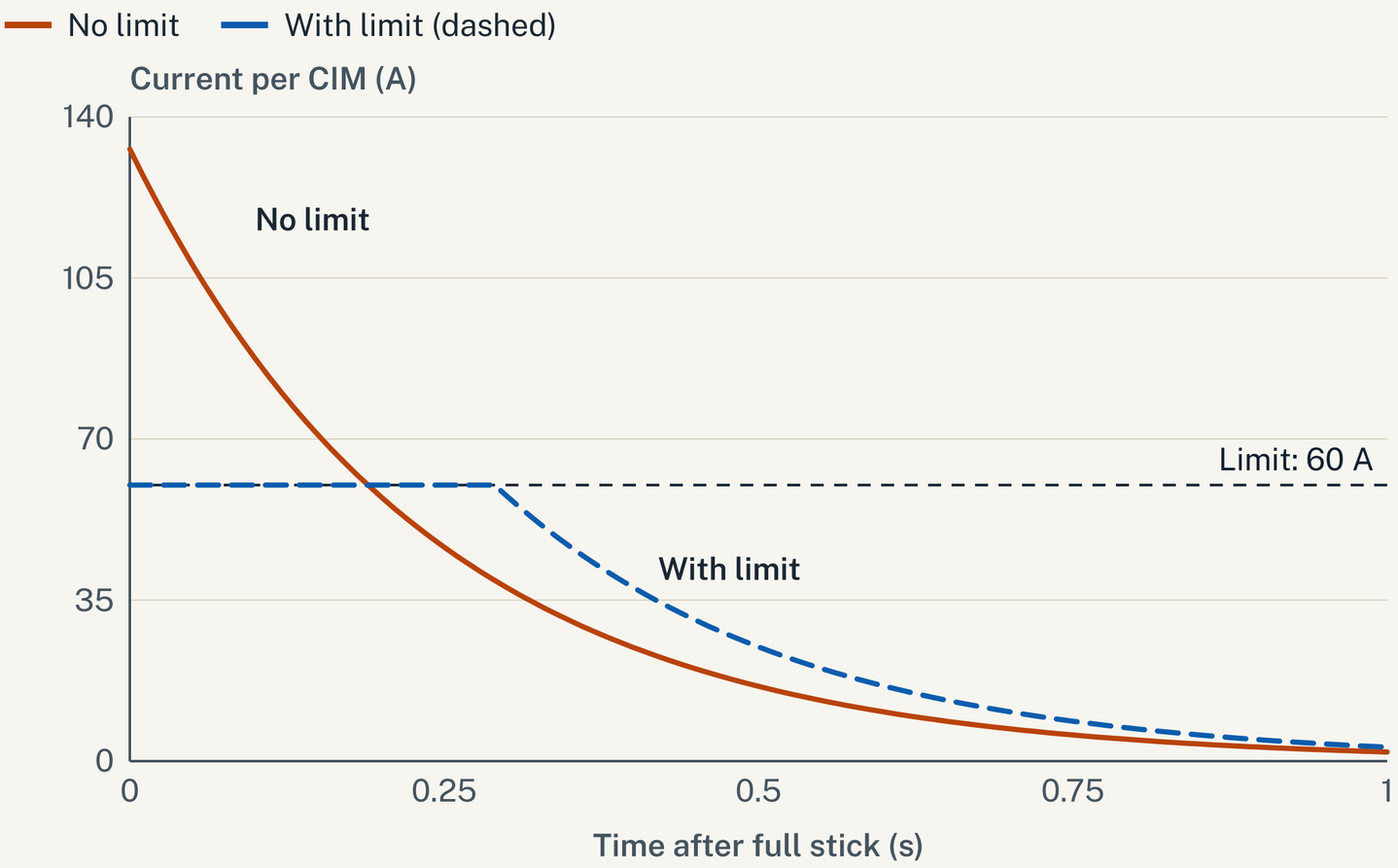
2. **Battery current**
   In the wires and breakers. It sags the battery.

3. **Same power, different split**
   12 V × 27 A in, 5.4 V × 60 A out. Same power.

4. **What the SPARK limits**
   Motor current. getOutputCurrent() reads it too.

CONTROL THEORY

# What a current limit protects




**1 The spike**
A stopped motor has no back-EMF, so the model starts at 133 A.

**2 What it protects**
Breakers, battery voltage and CIM heat.

**3 The cost**
3 m/s takes 0.32 s, not 0.21 s.

**4 Measure it**
This is a model. Log getOutputCurrent() during a short, hard push.

CURRENT

# Choosing a limit: stop where the wheels slip

DriveLimits.java

```
  /** Motor current where the wheels start to slip. */
  public static double slipAmps() {
1   double forceN = WHEEL_GRIP * ROBOT_MASS_KG * GRAVITY;
2   double wheelNm = forceN * WHEEL_RADIUS_METERS;
2   double motorNm = wheelNm / (DRIVE_MOTORS * GEARING);
3   return DCMotor.getCIM(1).getCurrent(motorNm);
  }

  public static SparkMaxConfig commonConfig() {
    SparkMaxConfig config = new SparkMaxConfig();
4   config.smartCurrentLimit(CURRENT_LIMIT_AMPS);
    config.idleMode(IdleMode.kBrake);
    return config;
  }
```

1. **Grip sets the force**
   Grip × weight. Our 1.0 and 50 kg are assumptions.

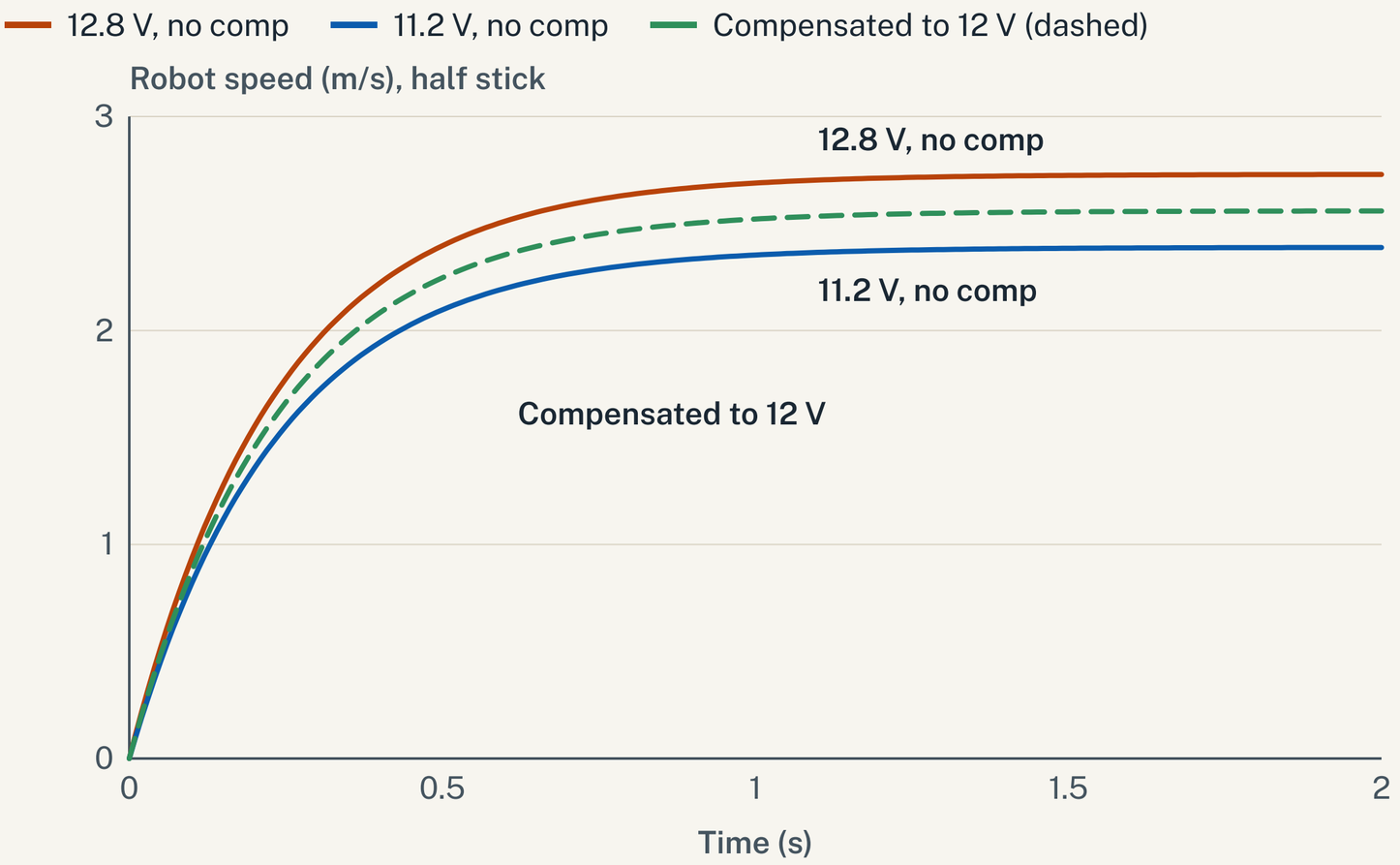
2. **Back through the gearing**
   Shared by 4 CIMs, divided by 8.45.

3. **Torque to amps**
   61 A per CIM. More just spins the wheels.

4. **Set the limit**
   FIRST’s 60 A is close to our number.

CONTROL THEORY

# Voltage compensation makes output repeatable




1. **Without it**
   set(0.5) means half of whatever the battery has right now.

2. **60 cm apart**
   That is the gap after 2 s between these two batteries.

3. **With it**
   The SPARK scales output to 12 V, so both batteries match.

4. **The cost**
   A little top speed on a full battery. It can’t add volts.

COMMON MISTAKE

# Comparing a motor limit with a PDH reading

| | |
|---|---|
| **What you see** | The motor is stalled at a 60 A limit, but its PDH channel shows about 27 A. One of them looks wrong. |
| **Why it happens** | getOutputCurrent() is motor current. The PDH measures battery current. At low speed they are far apart. |
| **The fix** | Compare getOutputCurrent() with your limit, and the PDH with your breaker. Log both. |

```
SmartDashboard.putNumber("Drive/MotorAmps", leftLeader.getOutputCurrent());
SmartDashboard.putNumber("Drive/AppliedOutput", leftLeader.getAppliedOutput());
SmartDashboard.putNumber("Drive/PdhChannelAmps", pdh.getCurrent(LEFT_LEADER_PDH_CHANNEL));
SmartDashboard.putNumber("Drive/BatteryVolts", RobotController.getBatteryVoltage());
```

BEST PRACTICE

# A limit for every motor

> **Give every motor its own limit, chosen for its job.**

REV suggests 40 to 60 A for a NEO and 20 to 40 A for a NEO 550. FIRST uses 60 A on KitBot CIMs. In REV’s stall test a NEO 550 lasted about 27 s at 40 A, and about 2 s at 80 A, the SPARK default.

WPILIB

# Checking your math with DCMotor

MotorCheck.java

```
public static void print() {
  // withReduction() applies the gearing to the motor's numbers.
  DCMotor side = DCMotor.getCIM(2).withReduction(GEARING);
  DCMotor cim = DCMotor.getCIM(1);

  double topSpeed = side.freeSpeedRadPerSec * WHEEL_RADIUS_METERS;
  double oneStalled = cim.getCurrent(0, NOMINAL_VOLTS);

  System.out.printf("Top speed, free-speed math: %.2f m/s%n", topSpeed);
  System.out.printf("Stall current: %.0f A per CIM, %.0f A for four%n",
      oneStalled, DRIVE_MOTORS * oneStalled);
}
```

withReduction() gears the motor down, so freeSpeedRadPerSec is now wheel speed. Radians per second × radius = metres per second.

EXERCISE

# Predict the KitBot, then check it

1. Work out top speed at **8.45:1** from the curve.
2. Now do **10.71:1**, another ratio WPILib lists for the KitBot chassis.
3. Write down stall current at 12 V for one CIM, then for all four.
4. Call MotorCheck.print() in RobotContainer, then run WPILib: Simulate Robot Code.
5. Set GEARING to 10.71 and run it again. Compare with your paper.

**Done when: your paper matches the console to two digits.**

RECAP

# What this deck covered

- **Voltage sets speed, current sets torque**, and the load decides the current.
- Free speed, free current, stall torque and stall current on any motor curve.
- Top speed by hand, and why peak power sits at half speed.
- Stall heat, battery sag and brownouts.
- Motor current versus battery current, choosing a limit, and checking it in logs.

**Next: Deck 9: driver feel, feedforward and where loops run.**

REFERENCE

# Official docs

**Brownouts and current draw**

docs.wpilib.org

**SPARK MAX current limits**

docs.revrobotics.com

**Locked rotor testing**

docs.revrobotics.com

**DCMotor Java docs**

github.wpilib.org